

Puneet Ludu

ML Engineer & Tech Lead

Production ML & LLM Reliability · 13+ years

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[Google Scholar](https://scholar.google.com/citations?user=...)

Experience

13+ years

Zillow (Zestimate), Machine Learning Engineer (Tech Lead), Sep 2021 – Present, Remote

4Y

Listing IQ: Interactive CMA Platform (Django, DocumentDB, PyTorch) [Real-time Valuations + Embeddings]

Led **proof-of-concept and core architecture** for platform now live as Listing IQ CMA: AI-powered tool enabling agents to customize property comparisons with map-based filtering, editable valuations, and amenity-level explanations. Designed APIs integrating real-time Zestimate valuations, property embeddings, and comparative analysis. Mentored junior engineer through production launch.

Impact: POC → shipped product powering agent pricing decisions; foundation for new agent revenue products

Zestimate Customer Care Agent + Reliability Infrastructure (React, LangChain, FastAPI) [RAG Grounding + Output Validation + Deflection KPI]

Designed and shipped the **first LLM assistant deployed to Zillow customer support**: personalized Zestimate explanations grounded in real valuation data. Built **RAG pipeline that constrains the agent** to answer from actual Zestimate features and values rather than free-form generation; layered **output validation with schema and range checks** to catch out-of-distribution responses before they reach customers. Deployed with **deflection rate** tracked as the primary production KPI.

Impact: First LLM tool deployed to Zillow customer care; production-hardened against hallucination via grounding and output validation

Active Listing Comps Engine (Apache Spark, Metaflow, H3 Geospatial) [Similarity Scoring + Batch/REST APIs]

Architected and built similar listings comparison engine from ground up: batch pipeline and real-time API using **H3 geospatial indexing** and customizable ranking algorithms. Processes **3M+ listings**, powers **Zillow Showcase** dashboards (**8M monthly views**) helping agents demonstrate listing performance vs non-Showcase homes. Led migration to append-only listing data source; built lifecycle management and deduplication layer to track active, sold, and removed listings across markets. **Production in 5 weeks**. Later led algorithm improvements.

Impact: 5 weeks to production; 95% complaint reduction, coverage 90%→98.5%; metrics cited in sales, marketing, and investor communications

Infrastructure & Engineering Leadership (Terraform, AWS, Metaflow, Docker)

Built Valuation API from POC to production: FastAPI service handling **~6K requests/day with zero P2 alerts** since launch. Led Zestimate 6.6 deprecation (Redis→hybrid cache feature store for real-time valuations, legacy system shutdown), saving **\$350K annually**. Drove several Zestimate model point releases to completion; created MR tracking system, led ETL optimizations. Completed CI/CD pipeline modernization across all team services with **zero production incidents**.

Impact: \$500K+ combined annual savings, 61% alert reduction (641→249 YoY)

Next-Gen Zestimate: Multimodal + NAM + Explainability (PyTorch, CLIP, Databricks) [DualLossAutoEncoder + NAM + Per-feature Attribution]

Leading Image Embeddings project: experimenting with DualLossAutoEncoder to predict property condition from **CLIP embeddings (3.3M listings)** for Zestimate integration. Prototyping a **NAM-based (Neural Additive Models) Zestimate variant** for intrinsic interpretability, alongside a per-feature attribution architecture that exposes dollar-level contributions to downstream AI agents. Led core Zestimate neural network ETL pipeline migration to Databricks; **identified and resolved 8x latency and 10x cost regression** before production deployment.

Impact: Building multimodal and explainability infrastructure for next-generation valuation products

Mentorship & Technical Leadership

Managed summer intern (2023): designed project plan, weekly check-ins, received **“strongly favorable”** feedback. Mentored **4+ engineers** across many projects, deployments, and onboardings. Created **RFC templates, Stacked MR best practices** etc. adopted by the team. Conduct technical interviews, lead code reviews, and coordinate cross-team design discussions.

Match Group (OkCupid), Machine Learning Engineer, May 2020 – Sep 2021, New York City

1.5Y

Discount Optimization (Python, Keras, TensorFlow, Weights and Biases) [Wide&Deep]

Owned **end-to-end ML pipeline** for subscription discount optimization: feature engineering, model training, A/B testing, deployment, and production monitoring. Discovered high prediction variance across model runs; designed **ensemble uncertainty estimation using 100-model bagging** to quantify and stabilize outputs for production deployment.

Impact: 6% overall revenue increase through A/B tested pricing models

 ML-Powered Financial Data Extraction (Python, TensorFlow, Keras, Sagemaker) [CNN, ELMo, BiLSTM]

Led multiple ML initiatives: (1) **Speaker identification** system for earnings calls using spectrograms and CNNs, (2) Private company fact extraction from **1.6M websites** using ELMo/BiLSTM. Rewrote MLangID language identification service. Led machine translation infrastructure (Polish SMT achieving **BLEU 69.10**).

Impact: 20% reduction in human-hours for earnings call processing, automated extraction from millions of documents 

 Financial Document Search & Ranking Systems (Apache Spark, Java, Python) [Distributed Trie, N-gram LM, Vector Space]

Led **team of 3 engineers** on Document Screening: built autosuggestion and concept similarity systems. Created FingerPrinter deduplication service (**10× response improvement: 1000ms→100ms**). Architected Formula Lookup using distributed trie and n-gram language models on Spark.

Impact: Improved formula ranking from 5.6 to 2.3, 66% faster document processing, powered StreetAccount trending news

 Technical Leadership

Established engineering best practices: Jenkins CI, comprehensive test suites, documentation standards. Mentored new hires and junior engineers. FingerPrinter became the model Java project within the ML group.

Tata Research Development and Design Centre, ML Research Engineer, Jul 2011 – Jul 2013, India

2Y

 Event Detection in Time Series (Java, Python, RapidMiner) [SVM - RBF]

Wrote an algorithm based on Shape Context for finding frequently occurring patterns and events, with as good results as SAX, DTW etc. with **7%** better results in the particular domain of car sensors.

 Data Harmonization Framework (DHF) (Java, Apache Pig)

Implemented an ETL framework that exploits the power of map-reduce and big-databases to fuse incongruous enterprise data from disparate sources in near real time.

Skills

Languages	Python · Java · C/C++ · Bash · SQL
LLM & GenAI	LangChain · RAG · LLM Guardrails · Embeddings · Vector DBs (Pinecone) · CLIP
Classical ML	PyTorch · TensorFlow · A/B Testing · Uncertainty Estimation · Model Monitoring
Data & Infra	PySpark · Databricks · MLflow · Metaflow · FastAPI · Django · Docker · Kubernetes · Terraform
Cloud & CI/CD	AWS (S3, EC2, SageMaker) · GitLab CI · Weights & Biases
Leadership	System Design · Technical Interviews · Mentoring · RFC Authorship · Cross-team Coordination

Publications

 [Google Scholar profile](#)[Inferring Latent Attributes of an Indian Twitter user using Celebrities and Class Influencers](#) ACM Hypertext 2015[Inferring gender of a Twitter user using celebrities it follows](#)

CORR 2014

[Architecture for Automated Tagging and Clustering of Song Files According to Mood](#)

IJCSI, 2010

Education

Master of Science in Computer Science, State University of New York, Buffalo, NY

2014

B. Tech. in Computer Science and Engineering, IIIT, India

2010

Open Source & Community

 Organizer @ MUFin

Program committee member, paper reviewer at top ML conferences: Workshop on Modeling Uncertainty in the Financial Sector (AAAI 2023, ECML-PKDD 2022)

 Lotion

Unofficial Notion.so Desktop app for Linux (2K+ GitHub stars / 60K+ Clones & Downloads)

 Romadeva

Tool to convert Roman script to Indic(Devanagari) script (Used by [Translators Without Borders](#))

 Quena

Question and Answering system – Indexed 1.6 Million Wikipedia documents, designed a question parser and a ranking algorithm based on popularity. (Apache Solr, NER, POS tagger)